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CHARACTERIZATION OF A LANTHANUM BROMIDE CRYSTAL

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Summary

The experience's aim is to characterize the detection efficiency of a Lanthanum Bromide (LaBr_3) scintillator crystal by studying its intrinsic radioactivity through the decay of ^{138}La . Then starting from this results, we want to identify an unknown radioactive source and estimate the activity of a radioactive source of Cesium (^{137}Cs).

The design of the machine is describe as show in figure 1

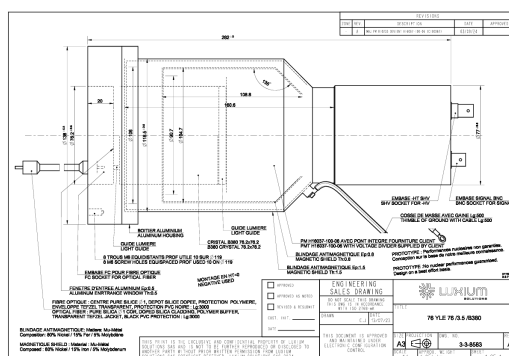


Figure 1: Design instrument

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Chapter 1

Instruments description

1.1 Tools list

- Phototube with scintillator crystal $\text{La}(\text{: Ce 5\%})\text{Br}_3$
- some radioactive source as ^{137}Cs , ^{22}Na and an unknown source
- voltage generator
- PhotoMultiplier Tube (PMT)
- BNC connections
- digitizer
- plates and supports of different sizes

1.2 PMT and scintillator crystal

The operation of a scintillator crystal is based on the process of radiation-matter interaction and its purpose is to transform the signal given by a photon into a measurable electrical signal. Following the interaction with an incident photon, an electron of the crystal is released, losing energy and exciting the electrons of the semiconductor, taking them from the valence band to the conduction band.

These, de-excited, emit an optical photon that cannot be absorbed by the semiconductor because it is doped. This photon produces, by photoelectric effect on the photocathode, an electron that is amplified inside a photomultiplier by a series of dynodes placed at different potentials, guaranteeing a gain of 10^5 or 10^6 electrons.

In our case, a crystal of Lanthanum Bromide doped at 5% with Cerium was used. Lanthanum Bromide contains a radioactive isotope, ^{138}La (a.i 0.08881%) with a half-life of $1.03 \cdot 10^{11}$ years that decays by electron capture (BR 65.2%) on an excited state of ^{138}Ba . When de-excited, it emits a γ beam at 1436 KeV.

The choice of this crystal was dictated precisely by its intrinsic radioactivity which allows to obtain important information on the detection efficiency of the instrument and allows to carry out its calibration. Furthermore, thanks to the short scintillation time, the crystal allows to have a high counting rate similar to what occurs in plasmas.

1.3 Digitizer, AC generator and BNC connections

About electrical components, in the experience we used an AC voltage generator connected via BNC cables to the instrumentation, thus powering the PMT inside the machine.

Instead, the digitalizer allows to convert the analogical signal we are evaluating into an electrical signal.

1.4 Support and plates

The last component are support and plates of different size. They are used mainly in study of efficiency at various distances. The size of plates are 5 or 10 cm.

Chapter 2

Data Analysis

First, connect the PMT to a high-voltage generator and, since you want to polarize the photo-tube negatively, it must be connected to the negative channel of the generator, while the positive channel of the latter is zeroed via a coaxial cable (which functions as ground).

The photo-tube is then connected to an oscilloscope and, by observing how the output signal varies, the voltage to use is determined. Even at low voltages, a signal begins to appear on the oscilloscope. This is because ^{138}La isotope, being radioactive, emits radiation, which, once converted into "visible" photons by the scintillator crystal, is detected by the PMT. After choosing the potential difference, equal to 1000 V.

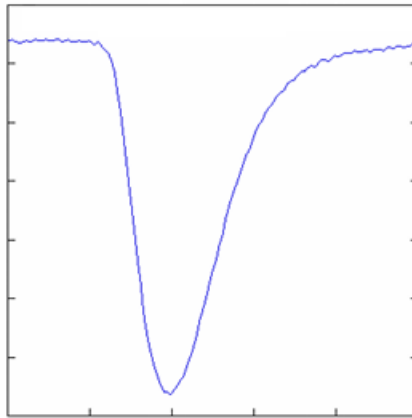


Figure 2.1: Oscilloscope signal

Note that, since the refractive coefficients of the scintillator crystal and the PMT lens differ, it is necessary to insert optical grease (whose coefficient has an intermediate value) between the two surfaces to avoid any total reflection phenomena. Looking at the real-time signal trend output from the photomultiplier tube on the oscilloscope, one notices the presence of some unexpected peaks probably due to the random arrival of a cosmic ray. Therefore, to avoid any type of external influence, the PMT was encapsulated inside the instrument and shielded by a metal cover.

We now proceed to the actual data collection: the output signal is connected to a digitizer that transforms the signal from analog (the one read on the oscilloscope) to digital. To calibrate the digitizer, we identified the three peaks of known energy, respectively: β decay at energy 788,74 keV and the double peaks of ϵ decay at energies 1440 and 1472 keV.

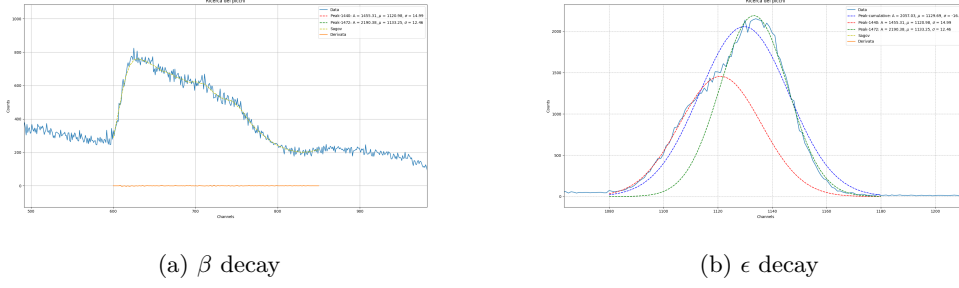


Figure 2.2: Calibration peaks

Finally performing a linear fit for the conversion from channels to energies, we can evaluate the calibration factor in according to the equation:

$$Energy[keV] = 1,29 \cdot Channel[Ch] + 4,47 \quad (2.1)$$

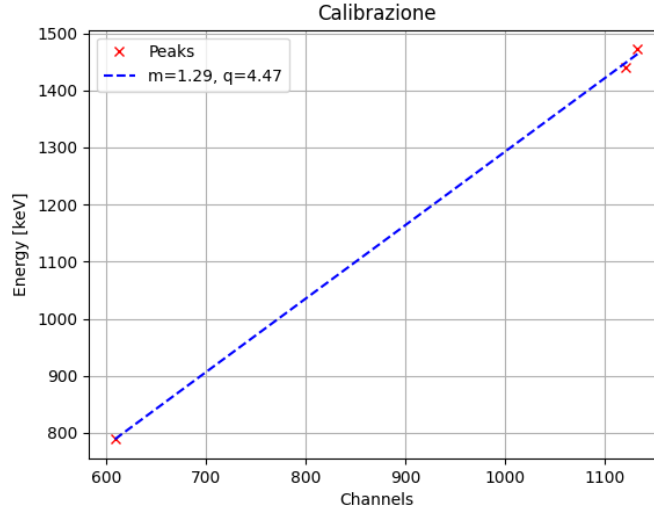


Figure 2.3: Calibration

Once the data needed to calibrate the crystal have been collected, the measurement can begin: after aligning the crystal and the cesium gel disk that will constitute the radioactive source on a workbench and verifying that the source and instrument are aligned with each other, data on the decay of the cesium are collected at various distances from the detector. In the two different cases analyzed (300 mm and 152 mm), it was necessary to lower the threshold to make the peak recognizable.

2.1 LaBr_3 scintillator

2.1.1 Intrinsic radioactivity

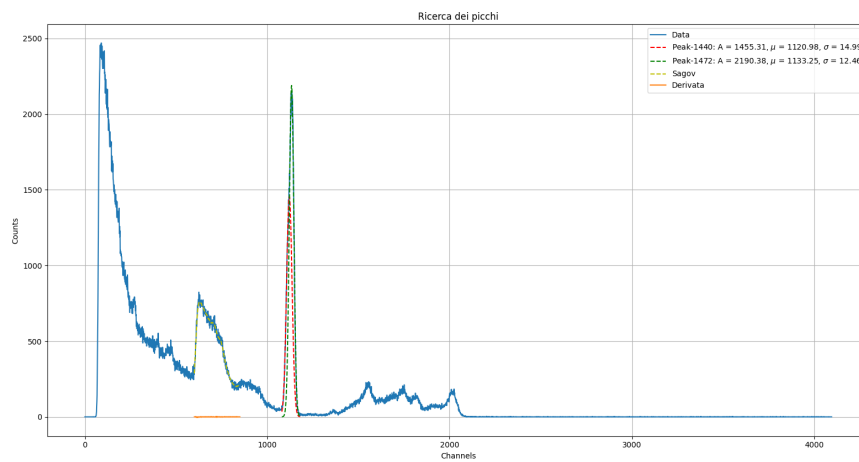


Figure 2.4: Background

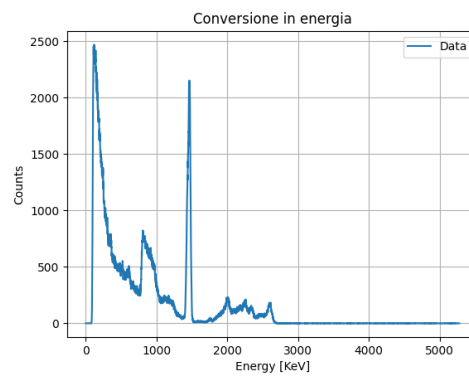


Figure 2.5: Background converted

2.1.2 Detection efficiency

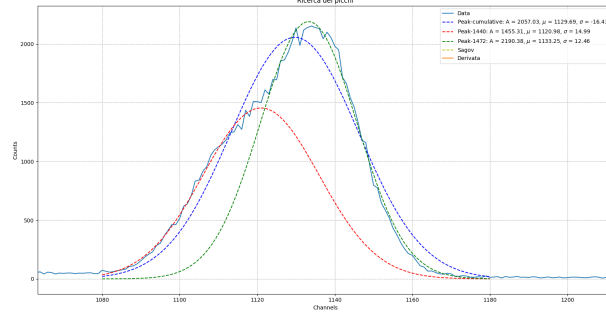


Figure 2.6: Double gaussian fit

$$R = \frac{FWHM}{H_0} = \frac{2.35\sqrt{FH_0}}{H_0} \quad (2.2)$$

errore sulla stima della centroide del fit totale: $1129 \pm 16Ch$
 la risoluzione data è del 3% , quella stimata invece è 6,99%, quindi il fattore di Fano sarà $\approx 0.18419937740610434$

$$\begin{aligned} V_{crystal} &= \frac{m_{LaBr_3}}{\rho_{LaBr_3}} + \frac{m_{CeBr_3}}{\rho_{CeBr_3}} = \\ &= \left(\frac{\tau_{1/2}}{\ln(2)t} \frac{\#}{BR\epsilon AI} \frac{1}{N_A} \right) \left(\frac{mm_{LaBr_3} 95\%}{\rho_{LaBr_3}} + \frac{mm_{CeBr_3} 5\%}{\rho_{CeBr_3}} \right) \end{aligned} \quad (2.3)$$

$$\begin{aligned} A_{LaBr_3} &= N_{AA}\lambda = \\ &= N_A \frac{(\rho_{LaBr_3} 95\% + \rho_{CeBr_3} 5\%) V_{crystal}}{(mm_{LaBr_3} 95\% + mm_{CeBr_3} 5\%)} AI \frac{\ln(2)}{\tau_{1/2}} \end{aligned} \quad (2.4)$$

$$\epsilon_{abs,RAD.INTR.} = \frac{\#}{A_{LaBr_3} t BR} \quad (2.5)$$

$$A_{Cs} = \frac{\#}{\epsilon_{ratio} \epsilon_{abs,RAD.INTR.} t} \quad (2.6)$$

$$A_{Cs} = A_0 e^{-\lambda t} = A_0 e^{-\frac{\ln(2)}{\tau_{1/2}} t} \quad (2.7)$$

2.2 ^{137}Cs source

2.2.1 Monte Carlo simulation

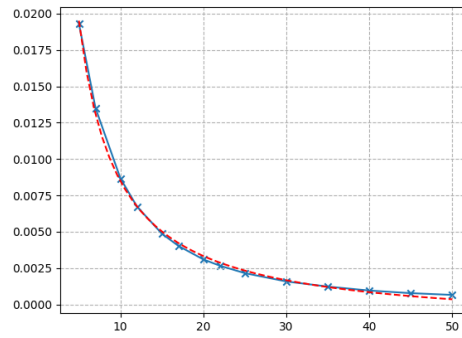
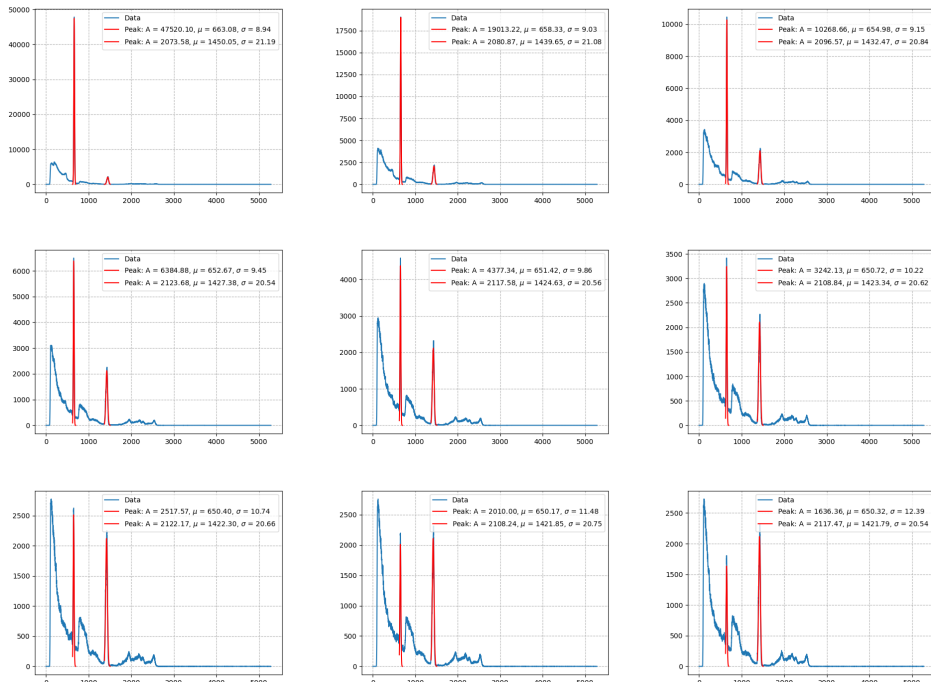


Figure 2.7: Simulation

2.2.2 ^{137}Cs measures

Appendix A

137 Cesium



Appendix B

Parameter

^{138}La half-life: $\tau_{1/2} = 1.02 \cdot 10^{11}$ years
 ^{137}Cs half-life: $\tau_{1/2} = 30.08$ years
Relative efficiency: $\epsilon = 19\%$
Branching ratio: $BR = 65.2\%$
Isotopic abundance: $AI = 0.08881\%$
Avogadro's number: $N_A = 6.023 \cdot 10^{23}$
 LaBr_3 molar mass: $mm_{\text{LaBr}_3} = 378.61$
 CeBr_3 molar mass: $mm_{\text{CeBr}_3} = 379.82$
 LaBr_3 density: $\rho_{\text{LaBr}_3} = 5.06\text{g}/\text{cm}^3$
 CeBr_3 density: $\rho_{\text{CeBr}_3} = 5.1\text{g}/\text{cm}^3$
Time: $t = 1200\text{s}$
 ^{137}Cs activity in 2015: $A_0 = 41.3\text{kBq}$

Appendix C

Error propagation

$$\sigma_F^2 = \sum_{i=1}^N \left(\frac{\delta F}{\delta x_i} \right)^2 \sigma_i^2 + 2 \sum_{i < j} \frac{\delta F}{\delta x_i} \frac{\delta F}{\delta x_j} \rho_{ij} \sigma_i \sigma_j \quad (\text{C.1})$$

σ_i represents the measurement error for the i-quantity, while ρ_{ij} is the correlation coefficient.